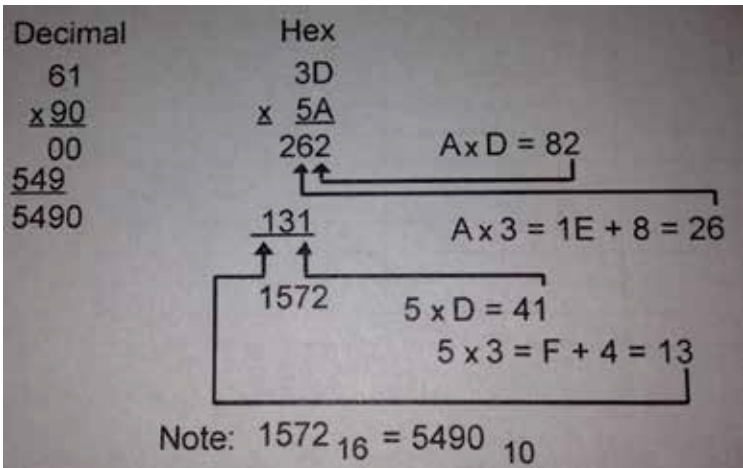


Registers and Hexadecimal Addresses

It's interesting to note that a single register can store any integer number between 0 and 4,294,967,295 on the Raspberry Pi. Translated to memory this only makes up 32 binary bits. While 0x20200000 may not appear to be a normal number it very much is, except that it is being written in the hexadecimal notation. The hexadecimal system recalculates the factors of numbers in base 16 and used alphabetic series from letter 'a' through letter 'f' to express six other unique values in hexadecimal form. We typically express numbers in base-10, for example 567 is actually 567₁₀ which is said as 567 to the base 10 which is the decimal system. In hexadecimal, 567 is written as 237 to the base 16 or 237₁₆ which can also be written as 0x237. This means that 0x20200000 is 20200000 to the base 16 or 538968064 to the base 10.

Our first command has been to place the number 0x20200000 in the r0 registry. This is the first basic step towards building our single purpose operating system. It is important to realise that computing involves allocating swaths of memory and devices using code. Each of these memory sets and devices has a specific address that is used to identify them. In the world of computers the numbers are just these addresses, and in the case of the Raspberry Pi the number 0x20200000 happens to be the address of the GPIO controller. This address location is decided by the makers of the Raspberry Pi and doesn't have any greater meaning or pattern. The



Understanding numerical expressions is a must to code detailed instructions.